

Constraint-Conditioned Actor- Critic for Offline Safe Reinforcement Learning

Offline Safe RL under Dynamic Safety Budgets

RL Mid-term Presentation

2026.06.03

Background: Why Offline RL?

Traditional reinforcement learning usually requires repeated interaction between an agent and the environment.

In many real-world domains, online exploration is expensive, risky, or unacceptable.

Robotics

Avoid collisions, falls, and unsafe motions

Autonomous Driving

Unsafe exploration cannot be used to collect data

Healthcare

Policies must satisfy strict risk budgets

From Offline RL to Offline Safe RL

Offline RL aims to learn a policy only from a fixed dataset, without additional environment interaction.

In safety-critical settings, the policy must also satisfy a cumulative cost constraint:

$$\max_{\pi} \mathbb{E}_{\pi} \left[\sum_t r_t \right] \quad \text{s.t.} \quad \mathbb{E}_{\pi} \left[\sum_t c_t \right] \leq \epsilon$$

where:

- r_t : task reward, such as speed, distance, or completion
- c_t : safety cost, such as collision, boundary violation, or overspeeding
- ϵ : a fixed safety threshold for the total acceptable risk

New Challenges in Offline Safe RL

1. Offline distribution shift

The dataset only covers historical behavior. At deployment, the learned policy may visit rare or unsupported state-action pairs, making value estimation unreliable.

2. Fixed safety constraints

Many Safe RL methods train for a fixed threshold ϵ . In deployment, the available safety budget may vary across scenarios, tasks, or time steps.

The core tension: maximize return while avoiding unsafe extrapolation and budget violation.

Our Research Question

Can we learn a policy from offline data that, at deployment time, can:

- Achieve high reward
- Satisfy safety constraints
- Adapt to different safety budgets
- Reduce the risk from OOD states and actions

$$\pi_{\theta}(a_t \mid s_t, \kappa_t)$$

In short: the policy should adapt its behavior according to the current remaining safety budget.

Core Idea: Condition on the Safety Budget

Instead of training a separate policy for each fixed threshold ϵ , we condition the policy on the current remaining budget:

$$\pi_{\theta}(a_t \mid s_t, \kappa_t)$$

Here, κ_t denotes the remaining safety budget at time step t . Given an initial budget κ_1 , it is updated after each step:

$$\kappa_{t+1} = \kappa_t - c_t$$

$$\underbrace{s_t}_{\text{state}}, \quad \underbrace{\kappa_t}_{\text{remaining safety budget}} \quad \longrightarrow \quad \underbrace{\pi_{\theta}(a_t \mid s_t, \kappa_t)}_{\text{constraint-conditioned actor}} \quad \longrightarrow \quad \underbrace{a_t}_{\text{action}}$$

$$a_t \quad \Longrightarrow \quad \begin{cases} R = \sum_{t=0}^T r_t, & \text{maximize reward return} \\ C = \sum_{t=0}^T c_t \leq \epsilon, & \text{satisfy total cost threshold} \end{cases}$$

Changing the initial budget κ_1 allows the same policy to trade off reward and cost adaptively.

Method: Constraint-Conditioned Actor-Critic

The training pipeline can be summarized in five modules:

Module	Role	Notation
Offline data	Build budget-annotated transitions	$D = \{(s_t, a_t, r_t, c_t, s_{t+1}, \kappa_t)\}$
Constraint-conditioned CVAE	Generate state-action pairs under a budget	$(\hat{s}, \hat{a}) \sim p_\phi(s, a \mid z, \kappa)$
OOD classifier	Detect whether (s, a, κ) is OOD / unsafe	$h_\psi(s, a \mid \kappa) \in [0, 1]$
Cost critic	Overestimate cost values for OOD samples	$Q_c(s, a \mid \kappa)$
Reward critic + Actor	Maximize reward under safety constraints	$Q_r(s, a \mid \kappa), \pi_\theta(a \mid s, \kappa)$

$$D \rightarrow p_\phi(s, a \mid z, \kappa) \rightarrow h_\psi(s, a \mid \kappa) \rightarrow Q_c(s, a \mid \kappa) \rightarrow \pi_\theta(a \mid s, \kappa)$$

Take-away: OOD / unsafe samples are assigned high cost, so the actor learns to avoid high-risk actions.

Feasibility and Experimental Plan

We plan to evaluate the framework on the DSRL benchmark.

Dimension	Plan
Environments	Run / Circle / Velocity-style safety control tasks
Metrics	Normalized reward, normalized cost, constraint violation
Baselines	BC-safe, CQL-based Safe RL, Lagrangian Offline RL, sequence modeling methods
Ablations	Remove CVAE, remove OOD classifier, fixed vs. varying safety budgets

Feasibility: the datasets and environments are available; Actor-Critic, CVAE, and classifier modules are standard; by the final presentation, we can first reproduce the main reward-cost trade-off and then add one extension such as dynamic budget switching or sparse safe trajectories.

Related Work

Direction	Methods	What they address	Limitation
Offline RL	CQL, IQL, BCQ, BEAR	Distribution shift	No explicit safety constraints
Safe RL	Lagrangian, CPO, PID-Lagrangian	Online constrained optimization	Needs interaction; often fixed thresholds
Offline Safe RL	CPQ, COptiDICE, VOCE, FISOR	Reward-cost trade-off from offline data	Hard to adapt to changing budgets
Sequence / trajectory	CDT, TREBI	Condition on reward / cost	Limited generalization to unseen budgets
CCAC	Our target framework	Budget conditioning + OOD detection	Depends on CVAE and classifier quality

Key gap: adapting behavior under dynamic safety budgets.

Q&A

Thank you!

References

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